

CANDIDATE'S RECEIPT

INSTRUCTIONS TO CANDIDATE

1. Fill in all the information requested clearly in capital letters.

TEST CODE

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SUBJECT

INFORMATION TECHNOLOGY – Paper 02

PROFICIENCY

GENERAL

REGISTRATION NUMBER

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FULL NAME

(BLOCK LETTERS)

SIGNATURE

DATE

2. Ensure that this slip is detached by the Supervisor or Invigilator and given to you when you hand in this booklet.
3. Keep it in a safe place until you have received your results.

INSTRUCTION TO SUPERVISOR/INVIGILATOR

Sign the declaration below, detach this slip and hand it to the candidate as his/her receipt for this booklet collected by you.

I hereby acknowledge receipt of the candidate's booklet for the examination stated above.

Signature

Supervisor/Invigilator

Date



CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

24 MAY 2024 (a.m.)



FILL IN ALL THE INFORMATION REQUESTED CLEARLY IN CAPITAL LETTERS.

TEST CODE

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SUBJECT

INFORMATION TECHNOLOGY – Paper 02

PROFICIENCY

GENERAL

REGISTRATION NUMBER

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SCHOOL/CENTRE NUMBER

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NAME OF SCHOOL/CENTRE

CANDIDATE'S FULL NAME (FIRST, MIDDLE, LAST)

DATE OF BIRTH

D	D	M	M	Y	Y	Y	Y
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SIGNATURE



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FORM TP 2024092



TEST CODE **01229020**

MAY/JUNE 2024

CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
5. **If you use the extra page(s), you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Answer ALL questions.

1. Connect Academy is an independent school that serves students at the secondary level. The school has embraced the use of technology to enhance the teaching and learning of course content.

- (a) Identify TWO types of computers that can be used by teachers and students during lessons.

Laptop and Tablet

(2 marks)

- (b) Name ONE hardware device that may be useful for presenting lessons in the classroom.

Projector

(1 mark)

- (c) The school uses a database to manage its data which is stored on its local network. Identify ONE other storage method by which the school can gain access to its data in the event that the data on its local storage gets lost or becomes corrupted.

Cloud Storage

(1 mark)



- (d) The following data was taken from the school's database.

STUDENT			
Student ID	Student	Age	Grade level
CACS0069	Lea, Tomas	18	12
CACS0123	Hutton, Esther	15	10
CACS0127	McKenzie, Marie	19	13
CACS0131	Upton, Kellie	16	10
CACS0203	Parr, Noreen	16	11
CACS0240	Palmer, Shelley	13	7
CACS0302	Hodges, Gaynor	14	9
CACS0352	Wills, Shona	14	9
CACS0422	Newman, Abbie	13	8
CACS1002	Thorp, Rosie	12	7

COURSE								
ID	Student ID	Math	English	Biology	Physics	Chemistry	Computer Science	Modern Languages
1	CACS0069	66	53	39	100	41	37	50
2	CACS0123	53	31	58	75	99	67	67
3	CACS0127	97	95	97	63	69	31	37
4	CACS0131	32	63	37	63	37	73	80
5	CACS0203	100	68	46	69	76	60	65
6	CACS0240	97	35	88	72	70	83	68
7	CACS0302	40	90	89	55	81	100	78
8	CACS0352	88	51	89	61	75	97	71
9	CACS0422	50	38	38	54	79	49	54
10	CACS1002	82	95	78	40	87	96	27

- (i) State the appropriate data types that would have been used in creating EACH of the following field names of the Student table.

Field name	Data type
Student ID	Text
Student	Text
Age	Integer
Grade level	Integer

(4 marks)

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- (ii) State the field that would be used as the primary key in the Student table.

StudentID

(1 mark)

- (iii) Identify the field in the Course table that can be used to link the Course table and the Student table **and** state the name of the key given to the field.

Student ID. Because this is a Foreign key

(2 marks)

- (iv) The following is the result of a query which was designed to display the course average of each student.

Student ID	Course Average
CACS0069	53.13
CACS0123	64.00
CACS0127	68.38
CACS0131	53.00
CACS0203	72.38
CACS0240	72.25
CACS0302	76.25
CACS0352	72.13
CACS0422	52.38
CACS1002	71.13

State the feature that was used in the query to generate the course average.

Calculated field

(1 mark)

- (v) When creating reports in a database, grouping can be very useful. Explain the purpose of grouping.

Grouping organizes records with similar values into sets, making it easier to perform summary calculations (like totals and averages) for each groups.

(2 marks)

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- (e) The school recently held security workshops for both staff and students, to heighten their awareness of security threats and practices.

- (i) Describe TWO ways by which staff can protect their computers and data from possible security threats.

Encryption: Scrambled sensitive data so it cant be only read with a specific key

Firewalls: Use software or hardware to block unauthorized access to the local network

(4 marks)

- (ii) State TWO personal security measures students should take when they are online.

Avoid sharing personal information

Use strong, unique passwords for different online accounts

(2 marks)

Total 20 marks

2. Impact Security Services provides a number of security options for its clients that includes security guards for personal and business purposes as well as monetary (cash) pickup. The company recently added computer and data security services along with a product line of security equipment.

(a) The following is an invoice that was sent to a customer from the sales department.

Impact Security Services 28 Tropical Boulevard Carib Town www.impactsecure.com admin@impactsecure.com		INVOICE	
Name / Address Family Genetics 34 Dunbar Road		Date	Invoice #
		1/6/2023	21073
	Delivery	Terms	Rep
	1/30/2023	50% deposit to be made 1 week before delivery	Ava Spence
Description	Qty	Cost	Total
Security camera	10	\$189.99	\$1,899.90
Alarm system	2	\$430.00	\$860.00
Security monitor	4	\$116.72	\$466.88
Biometric lock system	4	\$133.73	\$534.94
Labour - installation	1	\$580.65	\$580.65
Optional services	1	\$300.00	\$300.00
			\$0.00
		Subtotal	\$4,642.36
		Sales Tax (15.0%)	\$696.35
		Total	\$5,338.72

- (i) State the word processing feature that was used to create the invoice.

Table

(1 mark)

- (ii) Identify the justification that has been applied to all monetary values.

Right- Aligned

(1 mark)

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- (iii) The company's logo is missing from the invoice. The logo is stored in a desktop folder on the sales manager's computer.

List, in correct order, the steps that should be taken to insert the logo to the right of the company's name.

Click in the document to the right of the company name

Go to the insert tab and select pictures

Browse the desktop folder and select the logo file

Click insert and adjust the image wrapping to "In front of text" or "Square" to position it.

(4 marks)

- (b) The company's website was recently updated to include the new services and products available.

Outline TWO checks that should be carried out before publishing a website.

Link Testing: Verify that all internal and external hyperlinks lead to the correct destination

Cross-Browser Testing: Ensure the website displays and functions correctly on different browsers like Chrome, Safari, and Firefox

(4 marks)

- (c) When customers hire Impact Security Services for their computer and data security needs, the company does an audit to assess the areas of vulnerability and the possible threats that may occur.

Define EACH of the following terms in relation to computer security and provide an example of each.

- (i) Vulnerability

A weakness in a system/ network that can be exploited

Outdated antivirus software

- (ii) Threat

A potential danger that can be exploit a vulnerability

A hacker or A malware

(4 marks)

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- (d) The company hosts webinars on its website every month to discuss the different types of security threats and attacks that companies may be exposed to. State ONE example of EACH of the following security threats and attacks that may be discussed at the webinars.

- (i) Industrial espionage

Theft of trade secrets or intellectual property from a competitor

- (ii) Denial of service attack

Overwhelming a server with traffic to make a website unavailable to users

- (iii) Ransomware

Malware that encrypts files and demands payment for decryption key

- (iv) Data theft

Unauthorized copying or removal of confidential information

(4 marks)

- (e) State TWO ways in which phishing messages can be sent.

Email

SMS

(2 marks)

Total 20 marks

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3. Kickstart Investments offers a range of options to its clients, from low risk to high risk investments. Due to an increase in clientele, the decision was made to open a second office in another city to accommodate more of its clients.

- (a) Both office locations have their own Local Area Network (LAN). Identify the type of network that would be needed to connect both locations.

Wide Area Network (WAN)

(1 mark)

- (b) Differentiate between an 'intranet' and an 'extranet'.

Intranet is a private network that can only be used by the organization staff

Extranet is somewhat of a intranet but can be partially accessed by authorized outsiders

(Such as customers and venders)

(2 marks)

- (c) Complete the following paragraph by filling in the blank spaces with the MOST appropriate term provided from the word list below.

software developer

switch

web server

Internet

network adapter

modem

network administrator

web manager

server

web page

copper cable

hyperlinks

Shortly after the opening of the new office, some of the employees and customers were experiencing difficulties completing tasks. The network admin did some troubleshooting and realized that the modem that converts digital data to analog signals was disconnected from the switch that manages and serves all the computers on the network. This prevented the employees from going online and accessing the internet. The employees on the third floor were unable to access certain files on the network due to a power supply (hardware) failure of the copper cable that was used to connect all the devices on that floor to the main computer that manages the network. The web manager also did some checks which revealed that some of the hyperlinks on the website were broken. This prevented customers from completing certain transactions.

(7 marks)

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- (d) Kickstart Investments needs an algorithm written to accept the initial investment amount made and the rate of return. The algorithm should calculate and print the return on investment earned using the formula below.

*Return on investment = (initial investment + (initial investment*rate/100)) – initial investment*

- (i) Complete the following Input Processing Output (IPO) chart for the algorithm to be written.

Input	Processing	Output
initial invesments rate	Return on investments = (initial invesments * rate / 100)	Return of investments

(3 marks)

- (ii) Write the pseudocode for the algorithm.

```
READ initial_investment, rate
Return_on_investment = initial_investments * rate / 100
PRINT Return_on_investment
```

(8 marks)

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- (iii) If initial investment is greater than \$100 000, then the rate of return is 45% and if initial investment is less than or equal to \$100 000, then the rate of return is 37%.

Write an appropriate compound conditional statement to evaluate each rate.

IF initial investment > 100000 THEN rate = 45 ELSE rate = 37

(3 marks)

- (iv) State the return on investment that would be received if the initial investment amount was \$150 000 and the rate of return was 45%.

$(\$150,000 \times 45 / 100) = \$67,500$

(1 mark)

Total 25 marks

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4. Nigel is the manager at Elite Sports, a small sports club that provides its members with access to personal training and multiple sporting activities.

(a) Nigel takes the data from application forms filled out by potential customers and enters it into the system for effective record-keeping.

(i) State ONE type of error that could be present after entering the data from the application forms.

Error Type

(1 mark)

(ii) Describe TWO verification checks that can be performed when entering data from application forms.

Double Entry: Entering the same data twice and having the system compare them for matches.

Visual Check: Manually comparing the data on the screen with the original source document.

(4 marks)

(b) The following spreadsheet shows monies collected from members for training at the sports club for one week.

	A	B	C	D	E
1	Elite Sports				
2	Member Name	Activity	Session hours	Cost/hour	Amount due
3	Lisa Allen	Badminton	8	\$1,300.00	\$10,400.00
4	Philip Sanders	Badminton	6	\$1,300.00	\$7,800.00
5	Tina Davis	Basketball	12	\$1,100.00	\$13,200.00
6	Catherine Morgan	Basketball	9	\$1,100.00	\$9,900.00
7	Fred Young	Swimming	4	\$1,000.00	\$4,000.00
8	Jerry Watson	Tennis	12	\$1,500.00	\$18,000.00
9	Doris Hughes	Tennis	10	\$1,500.00	\$15,000.00
10	Carolyn Johnson	Tennis	6	\$1,500.00	\$9,000.00

(i) Name the spreadsheet function that will display the number of cells in the range B3:B10 that contains the activity "Tennis".

=COUNTIF(B3:B10, "Tennis")

(1 mark)

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- (ii) Write the formula that was used in Cell E3 to calculate the amount due.

=C3 * D3

(3 marks)

- (iii) The IF function performs a logical comparison between two values in a spreadsheet.

Describe the syntax of the IF function.

=IF(logical_test, value_if_true, value_if_false)

(4 marks)

- (iv) Identify both the primary field and the secondary field used to sort the data as displayed in the spreadsheet at (b).

Primary field Activity

Secondary field Member Name

(2 marks)

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- (c) The following pseudocode was created to generate the total funds received from member payments.

Start

total = 0

Print "Enter amount due for member"

Read amountdue

While (amountdue $\geq$ 0) do

total = total + amountdue

Print "Enter amount due for next member"

Read amountdue

Endwhile

Print "Total amount collected:", total

Stop

Design a flowchart to represent the pseudocode above.

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(10 marks)

Total 25 marks

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